Phase 4.25: Rank-5 Deferred Startup State

Tactix development report for Andrew

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1 Outcome

Rank 5 exposed a genuine startup-lifetime memory excess rather than a search or simplification defect. The accepted solver allocated several per-variable images for a 58,633,530-variable SAT formula even though early Lucky terminates before preprocessing or search can consume them. The promoted repair defers the cold rule state and DefaultPolicy guidance until the first failed early Lucky opportunity.

The decision-grade three-wave result is summarized below. All six SAT models were independently checked against the original CNF and every run left zero certificate and scratch residue.

Metric	Accepted Tactix	Repaired Tactix	Current Kissat
Median peak RSS (KiB)	14,368,044	10,075,752	10,402,844
Median wall time (s)	—	21.550	29.913

The repair removes 4,292,292 KiB, or 29.873878%, from accepted Tactix's observed peak. Repaired Tactix is 3.144256% below current Kissat in median peak RSS and retains the early-terminal time advantage at a 0.717080 paired median ratio. All three samples of each solver are within the required 2% stable-timing band.

2 Portable lifecycle boundary

The kernel can now defer 40 bytes per declared variable of BVE retry, touch epochs, Backbone priority, analysis marks, and probing stamps. The shipped DefaultPolicy frontend additionally leaves roughly 35 bytes per variable of VMTF, EVSIDS, phase guidance, and compact import-order storage unattached. Startup propagation and early Lucky still advance the same event counters and epochs while these images are absent.

If early Lucky returns unknown, a single explicit boundary reconstructs the first-round cold-state predicates. The existing `ScopedHeuristics` path then consumes the parser-retained activation order before sparse collection and every subsequent preprocessing or search rule. Reader helpers assert that the cold storage has materialized. Random, replay, stress, bare-kernel, and caller-supplied heuristic sessions retain their eager lifecycle.

This is an allocation-lifetime repair only. It changes no proof rule, calculus, rule admission, budgets, candidate order, logical transition, scheduling state, or N1–N4 disposition. It contains no benchmark identity, rank, hash, deadline, per-instance branch, or tuned threshold. Root values created before heuristic attachment are fixed and therefore outside the later decision surface.

3 Correctness and broad evidence

Release tests are 76/76 PASS and the event-trace build is 75/75 PASS. New tests cover early terminal execution with both images absent, exact materialization predicates, and eager-versus-deferred failed-

early execution with equal decisions, conflicts, propagations, and checked extended models. The Rank-4 candidate trace through 100 conflicts is byte-identical to the accepted 33,138-line trace.

The complete frozen B28+C5 gate is 66/66 PASS: 34 SAT models independently check and 32 UNSAT proofs independently verify. There is no solve loss, invalidity, cleanup failure, or certificate residue. Candidate/predecessor PAR-2 is 1,326.154/1,329.271 seconds, a 0.997655 ratio below the frozen 1.10 ceiling; peak memory is also neutral.

Claude Code exclusively owned the required fuzz campaign and issued the authoritative terminal PASS. It reports 81/81 corpus replays and 30,000/30,000 generated cases, including 16,041 independently checked SAT models, 11,035 native-LRAT-checked UNSAT results, and 27,076 differential verdict agreements. It reports zero crash, sanitizer, invariant, rejected model, rejected proof, differential, certificate, scratch, or cache failure. Direct reach covered early SAT and startup-root-refutation terminals while both images were absent, failed early paths crossing both materialization boundaries, and eager-lifecycle negative controls. No repair workflow was required.

4 Evidence authority

The isolated evidence is frozen under these campaign directories:

- `20260827-0950-tierd-rank5-memory-boundaries`: accepted baseline;
- `20260827-1010-tierd-rank5-cold-state`: cold-rule-state arm;
- `20260827-1040-tierd-rank5-deferred-guidance`: complete candidate;
- `20260827-1100-tierd-rank5-deferred-guidance-three-wave`: stable timing;
- `20260827-1110-BC-rank5-deferred-state-guidance-gate`: B28+C5 gate; and
- `20260827-1120-rank5-deferred-state-guidance-fuzz`: Claude-owned fuzz.

The repaired Release binary SHA-256 is `26430a9dfa2d...71f9977f34845`; the exact value is recorded in the technical report and frozen manifests. The solver-and-test diff SHA-256 is `c15d14e88431...86fa016f777b`.

5 Screen status and next action

The independent accepted-binary Rank 1–40 reevaluation is 79/80 terminal. Its sole remaining item is current Kissat Rank 18, whose solver result is UNSAT and whose widened independent proof check is still running. This does not block landing the isolated Rank-5 repair, but the campaign remains non-scorable and no witness will be selected until that checker exits and all losses are classified. The fresh screen has no Tactix OOM; its known Tactix solve timeouts are ranks 15, 16, 19, 23, and 36.

Rank 4 is independently valid at 10.733/7.728 seconds, with portable trace detail exact through 1,000 conflicts; its residual gap is downstream search throughput. Rank 32 is independently valid at 410.828/292.637 seconds and 582,204/299,988 KiB. After the screen becomes terminal, Rank 32 is the next declared target. The previously rejected Congruence-admission arm remains closed under N4; new work must localize a portable memory or downstream search/throughput owner before altering behavior.